Sample Access & Storage Section Draft (1-2 pages)

Project/System Name: (Example) Campus Lab File Server

Group Number: ____________________

List of Members: ________________________ ________________________

________________________ ________________________

Date: ____________________

A. Purpose

This section defines how storage is provisioned, controlled, and monitored to ensure secure and fair access for all users. It prevents one user or team from consuming all resources and supports reliable operations.

B. Storage Provisioning

We created a safe test storage stack inside a VM using loopback + LVM. A 2 GiB loopback image was attached as a virtual disk, then configured as: PV - VG - LV - XFS filesystem - mounted at /mnt/labfs.

Evidence included: lsblk -f, pvs, vgs, lvs, and mount output showing /mnt/labfs with quota options enabled.

C. Quota Policy

To prevent noisy-neighbor issues, our quota policy includes:

1) User quota (student accounts)

- Soft limit: 500 MB
- Hard limit: 600 MB
- Soft inode limit: 5,000 files
- Hard inode limit: 6,000 files

2) Project quota (shared team folders)

- Folder: /mnt/labfs/projA
- Hard limit: 1 GB
- Hard inode limit: 200,000 files

Reason: User quotas control individual usage, while project quotas control shared folders regardless of file ownership.

Evidence included: xfs_quota report -ugi and xfs_quota report -p.

D. Monitoring Plan

We monitor three key storage signals:

- Capacity (space): df -h /mnt/labfs (warn at 80%, critical at 90%)
- Inodes (file count availability): df -i /mnt/labfs (critical if <10% free)
- Disk I/O pressure (performance): iostat -xz and pidstat -d

We identify the largest consumers using: du -sh /mnt/labfs/* | sort -h | tail.

Evidence included: outputs of df -h, df -i, and du results.

E. Verification and One Fix Implemented

Issue found: quota behavior did not apply as expected because the test user's permissions were too restrictive.

Fix applied: we used ACL to grant the user access: setfacl -m u:student1:rwx /mnt/labfs.

Verification: quota reports were rechecked using xfs_quota report -ugi and confirmed limits were active and visible.

F. How This Supports Secure, Fair Access

By combining quotas with monitoring, the system ensures that storage is allocated fairly and performance remains stable for all users. Quotas prevent a single user or shared folder from consuming all resources, while monitoring helps detect early warning signs (space, inode exhaustion, I/O saturation) so administrators can take action before service disruption occurs.